

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology (B.Sc.I.T Semester V)**  
**Data Visualization with Power BI and Tableau**

## Practical – I

|                               |                                    |
|-------------------------------|------------------------------------|
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| Class:TYIT                    | Batch:1                            |
| Date of Assignment:24-07-2026 | Date/Time of Submission:24-07-2026 |

### 1. Objective

To gain hands-on experience connecting Tableau to a CSV dataset, navigating the user interface, distinguishing between dimensions and measures, manipulating continuous and discrete fields, creating essential visual charts (bar chart, line chart, map), and assembling them into an interactive dashboard.

### 2. Dataset & Environment Overview

- **Dataset Name:** Data Set- Inc5000 Company List\_2014.csv
- **Software Tool:** Tableau Desktop / Tableau Public
- **Workbook File:** Book2\_v2026.1.twb
- **Attributes Included:** company, industry, city, state\_l, metro, revenue, growth, rank, workers, yrs\_on\_list, \_resultNumber.

### 3. Step-by-Step Procedure

#### Task a: Connect to CSV Dataset & Explore Interface

1. **Connect to Data Source:**
  - Open Tableau Desktop.
  - Under the **To a File** pane on the start screen, select **Text File**.
  - Browse and select Data Set- Inc5000 Company List\_2014.csv.
2. **Data Source Workspace Setup:**
  - Verify the field headers (company, industry, revenue, growth, city, etc.) in the data preview window.
  - Click on **Sheet 1** at the bottom left to open the visualization workspace.
3. **Interface Exploration:**
  - **Data Pane (Left):** Displays data tables divided into **Tables/Fields** categorized as Dimensions (Blue) and Measures (Green).
  - **Shelves (Top/Center):** Columns, Rows, Pages, Filters, and Marks cards.
  - **Canvas (Center):** The main visual grid workspace.

## Task b: Identify & Differentiate Measures and Dimensions

### 1. Inspect Data Categorization:

- **Dimensions (Categorical / Qualitative data):** Located at the top of the Data pane.
  - *Examples:* Company, Industry, City, State L, Metro, Url.
- **Measures (Numerical / Quantitative data):** Located below dimensions, ready for aggregation (SUM, AVG, COUNT).
  - *Examples:* Growth, Rank, Revenue, Workers, Yrs On List.

### 2. Geographic Roles:

- Observe spatial icons next to City and State L, representing Tableau's automatically assigned Geographic Semantic Roles.

## Task c: Convert Fields Between Discrete and Continuous

### 1. Understanding Field Types:

- **Discrete (Blue pill):** Creates headers and distinct categories.
- **Continuous (Green pill):** Creates continuous axes.

### 2. Step-by-Step Testing & Observation:

- Drag **\_resultNumber** or Rank onto the **Columns** shelf.
- **Convert to Discrete:** Right-click Rank on the shelf and select **Discrete**. Observe how Tableau creates individual headers for every numeric value.
- **Convert to Continuous:** Right-click Rank and switch it back to **Continuous**. Observe how Tableau instantly updates the visualization to display a continuous axis line.

## Task d: Create & Customize a Basic Bar Chart (Sheet 2)

### 1. Build the Chart:

- Rename **Sheet 1** tab to **Sheet 2** (or open a new sheet).
- Drag the **Industry** dimension to the **Rows** shelf.
- Drag the **Revenue** measure to the **Columns** shelf (aggregates as **SUM(Revenue)**).

### 2. Customize & Polish:

- **Sorting:** Click the **Sort Descending** button on the top toolbar to order industries by highest revenue.
- **Labels:** Drag **SUM(Rank)** or **SUM(Revenue)** onto the **Label** mark card to display formatted values directly on the bar ends.
- Set Mark type to **Automatic** or **Bar**.

## Task e: Build a Line Chart for Time-Series / Trend Analysis (Sheet 3)

### 1. Build the Chart:

- Create a new sheet named **Sheet 3**.
- Drag **\_resultNumber** (or date field if available) to the **Columns** shelf as a continuous field.
- Drag **Growth** and **Rank** measures to the **Rows** shelf.

## 2. Multi-Measure Customization:

- Combine axes or use dual-axes to track how **Growth** and **Rank** progress sequentially across **\_resultNumber**.
- Add **Industry** to the **Rows** shelf to generate small multiples per industry category.
- Set Mark type to **Line** or **Automatic** to observe trend trajectories.

## Task f: Create a Geographic Map Visualization (Sheet 1)

### 1. Generate Map:

- Create or open **Sheet 1**.
- Double-click on the **City** dimension (or drag **City** to **Detail** on the Marks card). Tableau automatically generates **Latitude (generated)** on Rows and **Longitude (generated)** on Columns.

### 2. Customize Map Elements:

- Drag **Growth** (SUM) to the **Color** mark card to create a sequential gradient legend indicating company growth intensity by location.
- Adjust Map Options via **Map > Background Maps > Tableau**.

## Task g: Combine Charts into an Interactive Dashboard (Dashboard 1)

### 1. Create Dashboard:

- Click the **New Dashboard** icon at the bottom tab bar and name it **Dashboard 1**.
- Set Dashboard Size (e.g., Fixed Size **1000 x 800** or Automatic layout).

### 2. Arrange Views:

- Drag **Sheet 2** (Revenue Bar Chart) onto the left side.
- Drag **Sheet 1** (Map) to the top right quadrant.
- Drag **Sheet 3** (Trend Analysis) to the bottom right quadrant.

### 3. Add Interactivity & Legends:

- Position the color legend for **Growth** and the quantitative filter for **Rank** on the right sidebar layout flow.
- Click the **"Use as Filter"** filter icon on Sheet 2 or Sheet 1 to enable cross-sheet filtering upon clicking specific data points.

## 4. Observations & Key Insights

### 1. Revenue Leaders (Sheet 2):

- **IT Services** generates the highest overall revenue (~**19B / 19,00,405** scale), followed closely by **Business Products & Services** and **Advertising & Marketing**.
- Sectors like **Security**, **Travel & Hospitality**, and **Engineering** exhibit the lowest revenue footprints in this dataset.

### 2. Geographic Concentrations (Sheet 1):

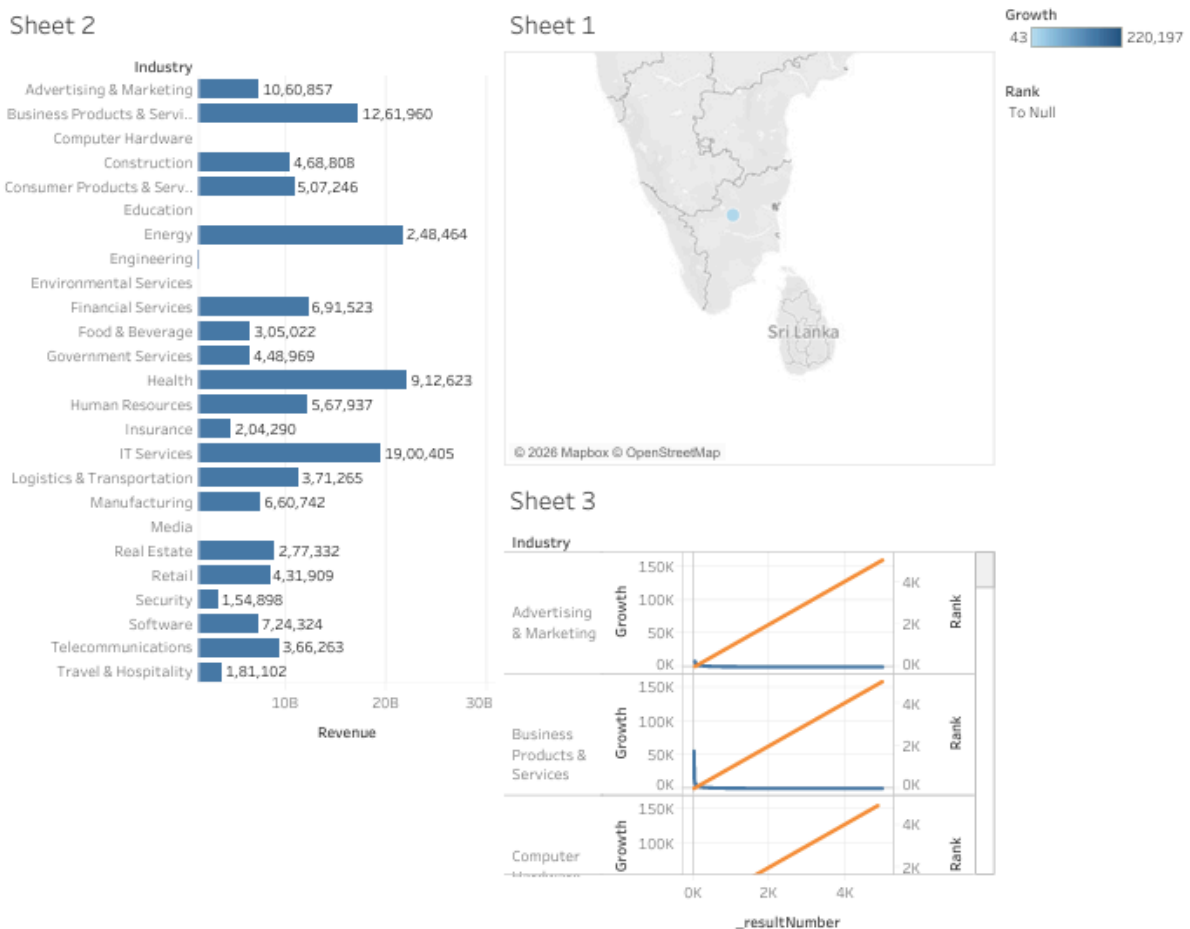
- High-growth companies show dense geographic clustering, revealing regional hubs for top-performing Inc 5000 businesses.

### 3. Discrete vs. Continuous Impact:

- Converting fields like **Rank** or **\_resultNumber** from discrete to continuous shifts the visualization focus from distinct headers/tables to trend-based continuous axes.

## 5. Deliverables Submission Checklist

- **Tableau Workbook File:** Saved as **Book2\_v2026.1.twb**.
- **Dataset File:** **Data Set- Inc5000 Company List\_2014.csv** attached in context.
- **Dashboard Layout:** **Dashboard 1** successfully combines Sheets 1, 2, and 3.
- **Lab Report:** Fully documented steps (a–g) with observations.



# Tableau Analytics Report: E-Commerce Performance & Dashboard Evaluation

## 1. Executive Summary

This report presents an analysis of the e-commerce sales dataset processed within Tableau workbook **bhavesh T039.twb**. The primary objective is to evaluate key operational metrics—including revenue generation, regional product performance, and customer satisfaction trends—while verifying whether the dashboard meets all technical submission criteria.

## 2. Dataset Overview: Dimensions & Measures

The data source (**ecommerce\_sales\_analytics\_5000.csv**) was successfully parsed into appropriate discrete and continuous fields:

| Dimensions (Categorical) | Measures (Quantitative) |
|--------------------------|-------------------------|
| Product Category         | Revenue                 |
| Region                   | Customer Rating         |
| Payment Method           | Quantity                |
| Order Date               | Unit Price              |
| Customer Id              | Discount                |
| Order Id                 | Delivery Days           |

## 3. Key Insights & Dashboard Observations

- Dominance of Electronics & Clothing:**  
The stacked bar chart in **product details** shows that **Electronics** and **Clothing** generate the largest share of total revenue across all four geographical regions (*East, North, South, West*), significantly outperforming **Beauty** and **Home** items.
- Payment Method Consistency:**  
Across all categories and regions, revenue distribution among various payment methods remains relatively uniform, indicating no single payment gateway bottleneck.

3. **Rating Dip Over Time:**

In the **product** worksheet, historical customer ratings tracked from 2026 through 2031 remain stable near upper thresholds before experiencing a steep sharp decline toward the end of the time horizon.

4. **Granular Customer Base:**

The customer ID breakdown shows a wide, evenly distributed volume of transactions without reliance on a small cluster of high-frequency buyers.

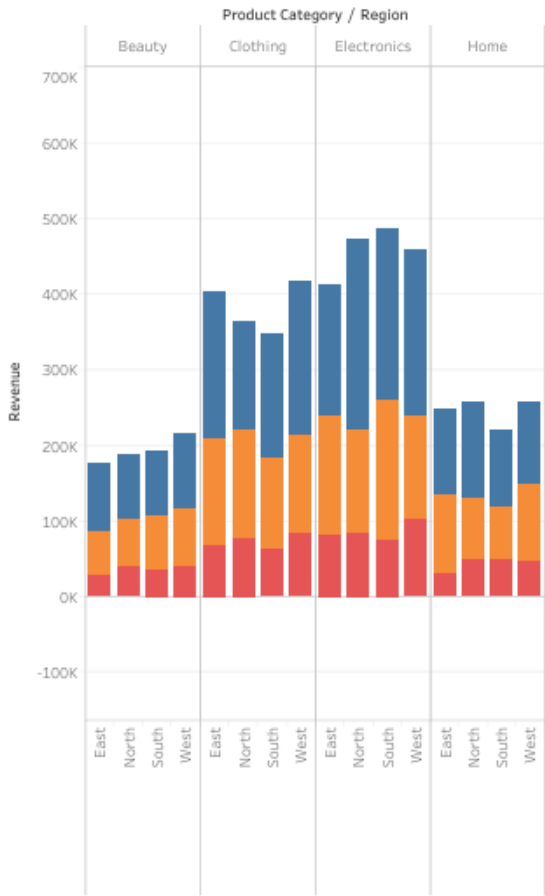
## 4. Dashboard Requirements Compliance Checklist

An evaluation of **bhavesh T039.twb** against the 6 assignment requirements:

| # | Requirement                       | Status     | Verification Details                                                                                                                                                |
|---|-----------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Import dataset into Tableau       | COMPLETED  | Dataset <b>ecommerce_sales_analytics_5000.csv</b> is correctly connected.                                                                                           |
| 2 | Identify Dimensions & Measures    | COMPLETED  | Proper categorization of dimensions ( <b>region</b> , <b>product_category</b> ) and measures ( <b>revenue</b> , <b>customer_rating</b> ).                           |
| 3 | Create 3 distinct visualizations  | NEEDS WORK | Currently contains only <b>2 worksheets</b> ( <b>product details</b> and <b>product</b> ). A 3rd unique visual (e.g., Pie Chart, Map, or Scatter Plot) is required. |
| 4 | Add interactive filter / element  | NEEDS WORK | Color legends are visible, but no interactive <b>Dashboard Action Filters</b> or dropdown controls are active.                                                      |
| 5 | Single dashboard with clear title | NEEDS WORK | Combined into <b>Dashboard 1</b> , but lacks a main text title banner at the top.                                                                                   |

|   |                                |           |                                       |
|---|--------------------------------|-----------|---------------------------------------|
| 6 | Write 3–5 insight observations | COMPLETED | Detailed in Section 3 of this report. |
|---|--------------------------------|-----------|---------------------------------------|

Revenue by Category & Region



Customer Rating Trends

